

POPULATION CONSIDERATIONS

UNDERSTANDING POPULATIONS

The differences in populations is an important concept to explore because it affects how you train yourself or your athletes if you are a coach. We have already discussed the difference in populations between beginner, intermediate, and advanced athletes in the section on the progression charts and levels. In this chapter, we are going to explore another four distinct categories of populations that encompass the vast majority of those who desire to pursue bodyweight training. Here are the different types of populations to consider:

- Emphasizing the differences in training between the sedentary and the active.
- Emphasizing the differences in training between the young and the old.
- Emphasizing the differences in training between those who are utilizing exercises for their sport and those who train for other reasons.
- Emphasizing considerations for those who are injured or recovering from an injury.

These categories are not mutually exclusive, so you may find that you or an athlete you train falls into more than one category. Considerations for constructing exercise routines must be made on all accounts to ensure that you safely progress in your training and address any problems that may arise along the way.

SEDENTARY VS. ACTIVE

The way that a sedentary person and the way that an active person approaches bodyweight training will naturally be different. To understand the differences in these two populations, let's further define each into different categories. The *sedentary* can be divided into two categories, while the active can be populations can be divided into five.

Sedentary

- Purely Sedentary
- Previously Athletic Sedentary

Active

- Non-Weight-Lifting Active
- Weight-Lifting Active
- Bodyweight Active
- Bodyweight Prepared
- Sport-Specific

The way in which you would want to introduce each of these populations into specific bodyweight training routines will differ based upon a number of factors. Let's outline some general principles for each of these populations for consideration in constructing your own routine and/or the routines of your clients. This is to ensure a suitable buffer for overuse injuries, and a solid introduction into bodyweight exercises based on previous history relative to training and particularly bodyweight training.

The *purely sedentary* are those who have never exercised in their life. There are a couple of factors to take into consideration with this population. First, you will want to start slowly. Often, the purely sedentary have very poor movement and body awareness, so attempting to progress too quickly could lead to injury. Those who fit into this population should pay special attention to learning correct techniques. This may mean slowing down movements by adjusting the tempo, or even not progressing at all if the movement is even slightly out of alignment.

The second factor with the purely sedentary population is that they have had no previous adaptations to exercise. Their bodies will therefore be adapting to the exercises as they perform them. This means that it is better for the purely sedentary to start with much easier exercises and higher repetitions. This will allow them to attain technical mastery, increase blood flow, and reduce the amount of stress on their slower adapting connective tissues. While higher intensity exercises will provide quick strength and muscle gain, one's joints and connective tissue will not be prepared for it, which can lead to overuse injuries. This will be a common thread you will see as you explore more populations.

Most purely sedentary will start off beginning in the L0-4ish range on the *Overcoming Gravity* charts, although most likely in the L0-2 range. The overall recommendation for this population would be to start with sets of five repetitions and work up to 12-20 repetitions in each set. The primary focus should be on technique, not merely moving fast through the exercises. After a few months of performing higher repetition work, you can progress without utilizing such high repetitions.

If a person is overweight or obese, place them in this category as well. In their case, they should place a priority on a healthy diet and consuming less calories, in order to lose weight. This is in addition to progressing through the charts much more slowly than those who fit in a more advanced population. It is common for those who are overweight or obese to have systemic chronic inflammation within their body, which negatively affects recovery and fat loss. Thus, losing weight in conjunction with taking exercise progression slowly will substantially relieve the burden on joints and connective tissue, which will ultimately decrease the potential for overuse injury.

The *previously athletic sedentary* are those who are former athletes or exercised regularly in the past. This category encompasses a wide range of athletes from different sports, both competitive and recreational. Previously athletic but sedentary would mean at least a three-month period or longer of inactivity. Attributes such

as endurance tend to start to degrade after three to five days of inactivity, whereas strength starts to degrade after about five to seven days. Of course, these numbers depend on various factors, such as the duration of training and if there is any overreaching or overtraining involved, which may elicit supercompensation. However, after three months it would be fair to say that there are significant decreases in overall ability from the athlete's previous performance level.

The first factor to keep in mind with the previously athletic sedentary is their minds are conditioned to hard work. Their work ethic is commendable and will be good for training in the long run; however, the first couple months back are the most dangerous for them. Because they are used to working intensely and their bodies previously made adaptations to exercise they will begin to progress extremely quickly. If too much exercise is performed too soon they can become exceedingly sore or potentially develop *rhabdomyolysis* (if the intensity is ramped up too quickly with certain types of exercises). You may see this occasionally in populations with intense exercise such as CrossFit, and even in bodyweight training if too much volume is introduced right away. If there is ever a massive amount of soreness coupled with increased localized swelling or edema, decreased ability to move through your range of motion, and your urine is brown, you should definitely go straight to the emergency room, as rhabdomyolysis is a life-threatening condition. Soreness with significant swelling and decreased movement through the range of motion without the brown urine indicates that a workout was severe, but not enough to elicit rhabdomyolysis. If any of this occurs, talk to a qualified medical professional. You may experience strength loss following any of these bouts, which will require you to ease into your routine very slowly. It is best to hold back for the first three to four weeks of the program for this population.

The second factor to keep in mind is your connective tissue integrity. The previously athletic sedentary population is the most at-risk for tendonitis and other connective tissue injuries because their strength and muscle mass will come back rapidly. This will lead to quick progress, but connective tissue strength and integrity will lag. It is imperative for this population to watch for signs of connective tissue compromise, such as twinges of pain, achiness, and generalized discomfort on the tendons, bones, and joints. When such symptoms occur, it is best to back off of the offending exercise(s) to a lower progression and work higher repetitions into the 15-20 (if not higher) range of not-to-failure exercise. The higher repetition work helps to push blood into the area for healing of the connective tissue. On the off chance that higher repetition work at a lower progression aggravates the situation, your best bet is to back off exercising that progression entirely and add in prehabilitation work (along the lines of isolation exercises and non-painful mobility work) to keep the affected areas moving and active. If the athlete is listening to their body correctly the vast majority of cases should resolve within a week or two, and then the athlete can slowly work their way back into exercise focusing on at least 12-15 repetitions to move to the next progression instead of lower amounts.

The *non-weight-lifting active* tend to be those in sports or recreational activities such as running, swimming, biking, and those who play sports recreationally without lifting weights or performing bodyweight exercises. For the most part, the same considerations for the previous athletic sedentary also apply to the non-weight-lifting active. In this population, there is also the illusion of the ability to progress quickly without incident because of the activity level. However, because this population is not accustomed to any type of strength training and has the drive to push themselves to be active, they may incur connective tissue injuries more rapidly than the purely sedentary. The same recommendation to remain aware of how the body is responding applies to this population.

For this population, another consideration may be the frequency of activity. If a person in this population is accustomed to running six times a week or playing their sport three to four times a week, this may have an impact on workouts. For example, their body is used to a high level of activity, which it can recover from at the present point in time. When an additional bodyweight routine of three times a week is added, their body may not respond as it should, especially if their recovery time is already stretched to the limit. This is the first population where there may have to be a balancing trade-off between how many sessions per week of sports and bodyweight workouts can be performed. The recommendation for a person in this population with a full schedule of activity is to start with two workouts a week and potentially reduce one to two days of other activities. Then, re-evaluate your level of fatigue and/or progress after two weeks. If you think your body can handle it, you can then add in another workout at this point or a bit later on.

The good thing about this population is that they will typically be both motivated and disciplined. Coaches can use this to teach proper technique and utilize higher repetitions if necessary in order to facilitate the movement patterns and ensure that strength is good. Strength training in this population can also be used to correct imbalances that may develop from training. For example, in climbers there is a tendency to have overdeveloped pulling muscles and backs compared to anterior muscles, such as the chest. Hence, balancing this out with additional pushing exercises, such as pushups and dips, will be effective in making sure one progresses quickly and safely with their discipline. This is true for any sport that may overemphasize strength, endurance, mobility, or flexibility in particular areas, at the expense of others.

The weight-lifting active are those who have experience lifting weights in a gym. There is a considerable difference between those who have lifted for one month, one year, and ten years. The more you have trained, the more structural adaptations your body has, which will have some transfer over to bodyweight exercises. For our purposes, if you have consistently trained for at least six months to a year, you fall into this category.

The weight-lifting active will transfer very well into most bodyweight exercise without any incident. The main categories of exercises that this population needs to worry about are the straight-arm strength exercises. These include isometrics, such as the planche, front lever, back lever, and the like, as well as straight-arm press handstands. Since this population tends to come in with a decent amount of applicable strength to bodyweight movements, the vast majority of exercises will come easily to them. Connective tissue strength with most of these will not be an issue. However, with the straight-arm exercises this strength is going to be detrimental to an extent because progression will happen much too quickly. Thus, one should continually monitor their body while performing exercises, particularly for signs of achiness, soreness, dullness, pinching, or stinging—especially around the joints, tendons, and ligaments.

Depending on how this population has trained before, they may have some imbalance issues if they primarily trained beach muscles. Thus, identifying and correcting any imbalances is important. For the most part, pushing tends to be approximately the same level or one grade higher than pulling on the *Overcoming Gravity* charts. If there are discrepancies in two or more levels it may indicate weaknesses that need to be shored up.

The *bodyweight active* are those who have experience with bodyweight exercises. For example, those who have trained with generalized bodyweight exercises such as pushups, dips, handstand pushups, pull-ups, rows, squats, pistols, and the like for at least six months. This population may have even experimented with more advanced exercises, such as one arm variations of pushups, pull-ups, or squats.

The main issue that the bodyweight active will run into is that they will have difficulty sticking to an organized strength routine. The vast major of athletes in this category are those who have previously focused on endurance or are haphazard in their training, randomly doing whatever they feel like. Their training is based on high repetitions and going to failure with many sets which is not how strength training should be structured. They will have difficulty staying consistent with a bodyweight strength routine because they feel like it is not doing much for them.

The *bodyweight prepared* are those who have experience with most of the straight-arm isometric strength exercises. Typically, this population will include those who are already engaging in structured bodyweight strength training. This includes gymnasts, b-boys, some parkour athletes, and others.

The main issue that this population has is their consistency, which is actually a good thing. However, if any overuse injuries do occur it can be a bad thing because they often do not want to reduce training in order to take care of their own bodies. In particular, those who are in competitive sports with bodyweight strength training will have the similar drive to ignore their injuries and continue training, lest they fall behind. Ignoring injuries or pre-injury conditions where there is achiness, tingling, and joint or connective tissue soreness is a recipe for developing long-term injuries. With this population, proper future time orientation and planning is important, so that they can rehabilitate themselves fully back to health and continue training at a high level.

In general, the bodyweight active and bodyweight prepared will already have adaptations to bodyweight exercise. The main concern with these populations is staying consistent with strength training and identifying any issues as they appear.

The *sport-specific* are those who play a sport competitively. They may or may not already have strength training based on bodyweight exercises or weights involved with their training regimen. This population is best dealt with on a case-by-case basis; it is always best to consult your coach if you have one to see what type of strength exercises, conditioning exercises, or other additional activity they recommend (if any at all).

Some programs—such as football and basketball—are more inclined toward lifting weights. In this case, refer to the *weight-lifting active* sports section for more details on what you should expect. In other cases, programs such as wrestling, gymnastics, or the military may use bodyweight exercises more extensively. If this is the case, refer to the *bodyweight active* section. The concept that you should take away from this section is not that there is a large difference in this population from the weight lifting active or the bodyweight active, but that these populations need to be more aware of how they use their time in training and recovery. If an athlete is training for and playing a sport, there needs to be an awareness of balance between strength and conditioning and the sport-specific training. This will be discussed later in the cross-training chapter.

YOUNG VS. OLD

The main difference between the young and the old populations is the decreased resiliency of the older population, which may be dealing with long-term issues such as decreased flexibility and mobility, previous chronic injuries, being overweight or obese, disease conditions such as diabetes or hypertension, decreased movement quality or inability to perform particular movements, and other functional limitations. For this population, it will be key to prepare the body before working out, especially if one cannot properly perform the techniques. In this case, modifications in workout structure will need to be made.

One phenomena that has been sweeping through the fitness world over the past decade is the awareness that, in general, static stretching should not be used prior to a workout, as it may decrease the ability to maximally contract muscles during strength workouts. However, there may be cases when static stretching prior to a workout may be beneficial if a gain in mobility is needed. This would be true if an athlete is having trouble moving into the right position or correctly performing a technique. Generally speaking, it is a good idea to place an emphasis on safety over performance when constructing a workout routine.

Naturally, the bodies of older folks are going to be less efficient overall in all phases of working out. This can be mitigated to an extent by a good warm-up. Spend ten to fifteen minutes warming up, and if you need extra time add your age divided by four. This will ensure sufficient time to allow the body to become more acclimated to your exercises, which will decrease your risk of injury. Warm-ups for various age groups may look something like this:

- 20-year-old: 10 minutes + $(20/4) = 15$ minutes
- 40-year-old: 10 minutes + $(40/4) = 20$ minutes
- 60-year-old: 10 minutes + $(60/4) = 25$ minutes

For younger athletes—especially those in the single digit ages—a primary concern will be decreased coordination, as the neurological pathways are not fully developed. This population also has the tendency to cheat their technique to pump out more repetitions. They think that more is better and that they are invincible. Clearly, this is unsafe when performing more advanced bodyweight movements where technique is paramount. If you are coaching or supervising younger children it would be a good idea to make a game out of exercise. Pair up the younger children and tell them what not to do. Children like to point out what other children do wrong and keep them in line, so it is basically a double-check structure for you. Use your resources to your advantage.

Older populations with limitations in flexibility and mobility will often have to add more of this type of training while simultaneously decreasing the amount of strength work due to their decreased capacity for training. The mobility and flexibility sections may start to look longer than the strength section. This is to be expected. A foundation of good movement and range of motion needs to be built before adding strength on top of it. Failing to do this is one of the easiest ways to get injured. Additionally, a full-body routine at a frequency of 3x per week may not be ideal for those who are older. They may need to work a split routine instead if they simply cannot recover from a large workout. For those who are older it is better to spread out a lot of work between various days of the week. Otherwise, an intense workout with everything in one day may rapidly lead to excessive fatigue and injury.

SPORT-SPECIFIC VS. RECREATIONAL TRAINING

The major difference between sport-specific and recreational training is the emphasis of the training. In sport-specific training, strength and conditioning and workout routines are only a means to an end. In other words, the primary goal is on-field performance and the strength and conditioning is supplemental to help improve on-field performance. This is different from training for recreational or other reasons, where the emphasis could be equally split between recreational sports and training, or there could be a greater emphasis on training and not as much for on-field performance.

In the previous chapter, we discussed attribute training. Sport-specific strength and conditioning is always going to be focused on attribute training, which will vary based on the sport you are playing. For instance, if you are a long-distance runner, much of your in-gym strength and conditioning will focus on endurance work, with a bit of power, strength, and plyometrics. On the other hand, if you are a sprinter for track and field, much of your in-gym strength and conditioning will be focused toward maximal strength, power, and plyometrics, with very little endurance training. This is where energy systems need to be taken into account.

Here are a few examples. One of the big differences between gymnastics and parkour is a lack of competition, although that may be slowly changing. In gymnastics, there is a defined season where you are competing and where you are not competing. This means that there are phases of in-season training and phases of off-season training. The vast majority of sports are seasonal—football, basketball, soccer, track and field, and the like. In-season training is typically approached differently than off-season training in that in-season training is meant to prepare the athlete to compete and maintain abilities. On the other hand, off-season training is typically used to increase the various abilities of the athlete so that they will come back harder, better, faster, stronger. *Daft Punk* reference intended.

The other major difference between sport-specific and recreational training is exercise selection. In sport-specific training, you rarely choose exercises you enjoy doing. You are required to perform the most effective exercises that help with the particular attributes needed for your sport. For example, squatting and deadlifting with free weights are typically used by athletic programs for strengthening an athlete. Exercises that would probably not be used are other variations of squatting, such as leg press machines and sissy squats (heels come off the ground, knees far forward). Some programs do not even like athletes who are making millions of dollars doing any explosive exercises such as power cleans (where the bar is pulled from the ground to chest level in one movement) because of the risk of injury. Basically, if you are the strength and conditioning coach and you injure a million-dollar athlete, you are probably going to get fired. Effectiveness of exercises is often mixed with safety of exercises to ensure that athletes are actually going to make it to the field on game day.

If the focus of your training is geared toward random recreational activities, developing awesome movements, or simply becoming strong, you can basically pick and choose what you want to learn and develop. This book lays out some general principles—such as structural balance guidelines—which will help you avoid paths that lead to injury. However, the rest of your fitness journey is up to you as you determine your unique goals and move toward them.

UNINJURED VS. INJURED

This population focuses on considerations for those who are injured vs. those who are coming back from an injury. There are two main distinctions here: 1) Those who are currently injured; 2) Those who have previously been injured, but have completed rehabilitation. Those who are injured will need to structure their workouts differently than those who are not injured. Some allusions to this concept have already been made in the attribute training section of Chapter 5, but we will expound heavily on this concept in the health and injury management chapter. Essentially, with injuries the goal is mixed between ensuring that the injured body part(s) are healing and becoming stronger again and ensuring safety in the programming and consideration to the surrounding areas.

There are also work-around methods to continue training while injured, so an injury should never stop you from training. One thing that keeps athletes sidelined is the belief that they cannot workout while they are injured. That is far from the case, and this mindset often leads to an athlete ceasing to workout altogether, which causes them to become sedentary. In this case, their diet typically will not change, which can cause them to become overweight or even obese. So you have someone who was physically active and in good shape who goes to sedentary and obese, and they look back and wondered how it happened. The injury was only a catalyst to a chain of poor decision-making. This is something that must be addressed with those who are injured. They must understand that even though something bad may happen during training, it is not a good reason to start making additional poor decisions that lead to slacking off from training.

For those who are rehabilitated from an injury, a generally good rule of thumb is to continue to do rehabilitation exercises at the end of a routine. This ensures continued activity to the body part in question, which will help build resilience against further or additional injury. It also keeps one aware that this body part was injured in the past, so special care for it should be taken in the future. The number one predictor of your next injury is a previous injury. You are much more likely to re-injure a body part than you are to develop a new injury altogether.

A special focus on prehabilitation after rehabilitation will potentially limit the amount of compensation patterns that can develop from an injury that appears to be fully healed. Compensation patterns often lead to overuse of other body parts. For example, if you suffer a knee injury in your right leg, you may start to favor more toward your left leg when squatting. Favoring the left leg may lead to a muscle imbalance or tendonitis of the hip or knee.

For those coming back from an injury, prehabilitation *after* rehabilitation for a certain amount of time is recommended. At minimum, it should be half as long as the period of rehabilitation; at maximum, it should be the full length of the rehabilitation, unless miscellaneous issues begin to arise and the period needs to be extended. Therefore, if it took two months to come back from tendonitis, you should be doing prehabilitation for 4-8 weeks after your rehabilitation period has ended. This will typically look like a couple of additional exercises at the end of your routine. While it is not much, it helps keep you focused on the reality that you were injured, so that you proceed with caution and avoid another injury from overuse. If miscellaneous issues (such as compensation patterns or other injuries) begin to occur, you may need to continue prehabilitation indefinitely. As always, if you are confused or have any specific questions regarding the implementation or rehabilitative or prehabilitative measures into your routine, you should talk to your orthopedic doctor or physical therapist.

CHAPTER 6 SUMMARY

POPULATION CONSIDERATIONS

Knowledge Base

Modifications based on various populations are important because these differences may drastically affect how you train. It is important to realize where you are and adapt your workouts accordingly, in order to prevent potential overuse injuries. Consult a medical professional before implementing a workout routine.

Here are the different populations we discussed:

Sedentary

- Purely Sedentary
- Previously Athletic Sedentary

Active

- Non-Weight-Lifting Active
- Weight-Lifting Active
- Bodyweight Active
- Bodyweight Prepared
- Sport-Specific

Application

If you fall into any of these categories, pay attention to the details that have been laid out in this chapter, as each group of individuals responds to bodyweight training differently. One group may have certain issues that require modifications, while another group may not. This must be taken into account when constructing a workout routine.

